Supply Chain & Inventory Analytics

Project Overview

This case study presents a **Supply Chain Analysis** project developed using **SQL, Microsoft Excel, and Power BI**. The project analyzes **10,000 supply chain orders** and inventory records to evaluate warehouse performance, inventory management, supplier efficiency, delivery performance, and order fulfillment.

The analysis focuses on optimizing inventory levels, monitoring warehouse revenue, evaluating supplier performance, tracking delivery metrics, and building interactive dashboards that support data-driven decision-making across supply chain operations.

Business Problem

The company experienced challenges in maintaining optimal inventory levels, managing supplier performance, and ensuring timely order deliveries across multiple warehouses. Overstocking increased inventory costs, while stock shortages affected order fulfillment and customer satisfaction. Management needed analytical insights to optimize inventory planning, evaluate warehouse efficiency, monitor supplier performance, and improve overall supply chain operations.

Stakeholders

- Supply Chain Manager
- Warehouse Manager
- Inventory Manager
- Procurement Team
- Logistics & Operations Team
- Senior Management

These are the people who would actually use the insights from your dashboards and reports

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Analysis Objective

- Optimize inventory management by identifying overstocked and understocked products.
- Evaluate warehouse performance to improve operational efficiency.
- Analyze supplier performance based on delivery time and order fulfillment.
- Monitor delivery performance to identify delays and improve customer satisfaction.
- Track key supply chain KPIs, including inventory value, order status, warehouse revenue, and delivery metrics.
- Develop interactive dashboards to support inventory planning and operational decision-making.

Dataset Description

This project uses **two related datasets** to analyze the complete supply chain process, from customer orders to inventory management.

The supply chain dataset consists of 10,000 order records integrated across five related tables: Orders, Inventory, Suppliers, Warehouses, and Customers. The dataset captures key supply chain operations, including inventory management, supplier details, warehouse performance, customer deliveries, and order fulfillment.

Key Attributes

- Order_ID
- Product_ID
- Warehouse_ID
- Supplier_ID
- Customer_ID
- Inventory_Value

- Stock_Available
- Reorder_Level
- Delivery_Days
- Order_Status
- Revenue
- Warehouse_Location
- Supplier_Name
- Customer_City
- Order_Date

Dataset Summary

- **Orders Table:** ~10,000+ records
- **Inventory Table:** ~800+ records
- **Relationship:** Product_ID
- **Tools Used:** SQL, Excel, Power BI

Tools & Technologies Used

- **Database:** MySQL – Data storage and business query analysis.
- **SQL:** Joins, Aggregate Functions, CASE Statements, Window Functions, CTEs, ABC Inventory Analysis, Inventory Replenishment Analysis, Supplier Performance Analysis.
- **Data Handling:** CSV Files – Data import, validation, and preprocessing.
- **Microsoft Excel:** Data Cleaning, Pivot Tables, Pivot Charts, KPI Reporting, Business Reporting.
- **Power BI:** Data Modeling, DAX Measures, KPI Cards, Interactive Dashboards, Data Visualization, Business Intelligence Reporting.

Analysis Performed

The supply chain dataset was analyzed using **SQL, Microsoft Excel, and Power BI** to evaluate inventory management, warehouse operations, supplier performance, and order fulfillment.

1. Warehouse Performance Analysis

- Ranked warehouses based on total revenue using SQL Window Functions.
- Compared warehouse revenue and inventory value.
- Identified top-performing warehouses.

2. Inventory Analysis

- Performed Inventory Replenishment Analysis to identify products below the reorder level.
- Conducted ABC Inventory Analysis to classify products based on inventory value.
- Evaluated inventory distribution across warehouses.

3. Supplier Performance Analysis

- Analyzed supplier contribution based on inventory value.
- Compared supplier performance to identify major inventory providers.

4. Delivery & Order Analysis

- Evaluated average delivery days across customer cities.
- Analyzed order status distribution (Pending, Shipped, Delivered, Cancelled).
- Monitored order fulfillment performance and delivery efficiency.

5. Dashboard & KPI Analysis

- Developed KPI reports and interactive dashboards using Excel and Power BI.
- Monitored Revenue, Inventory Value, Orders, Delivery Days, and Warehouse Performance.
- Visualized category-wise revenue, warehouse performance, order status, and delivery metrics.

Key Observations & Business Relevance

Key Observations

- Warehouse W03 generated the highest revenue among all warehouses.
- Office Supplies contributed the highest overall revenue.

- ABC Inventory Analysis identified high-value products requiring close inventory monitoring.
- Inventory Replenishment Analysis highlighted products that required immediate restocking.
- Supplier performance varied significantly based on inventory contribution.
- Average delivery time remained around six days across most customer locations.
- Order status analysis provided visibility into pending, shipped, delivered, and cancelled orders.

Business Relevance

The analysis provides valuable insights that support supply chain operations by:

- Optimizing inventory planning and replenishment decisions.
- Improving warehouse utilization and operational efficiency.
- Monitoring supplier performance to strengthen procurement strategies.
- Tracking delivery performance to improve customer service.
- Supporting management with KPI-driven reports and interactive dashboards for operational decision-making.

Supply Chain & Inventory Analytics Project | SQL, Excel, Power BI

- Analyzed **10,000 supply chain orders** covering inventory, warehouses, suppliers, and deliveries.
- Used SQL (Joins, CTEs, Window Functions, ABC Analysis) to generate business insights.
- Built Excel KPI reports and Pivot Tables to analyze warehouse and inventory performance.
- Developed an interactive Power BI dashboard to monitor key supply chain KPIs.
- Generated insights to improve inventory planning, warehouse efficiency, and delivery performance

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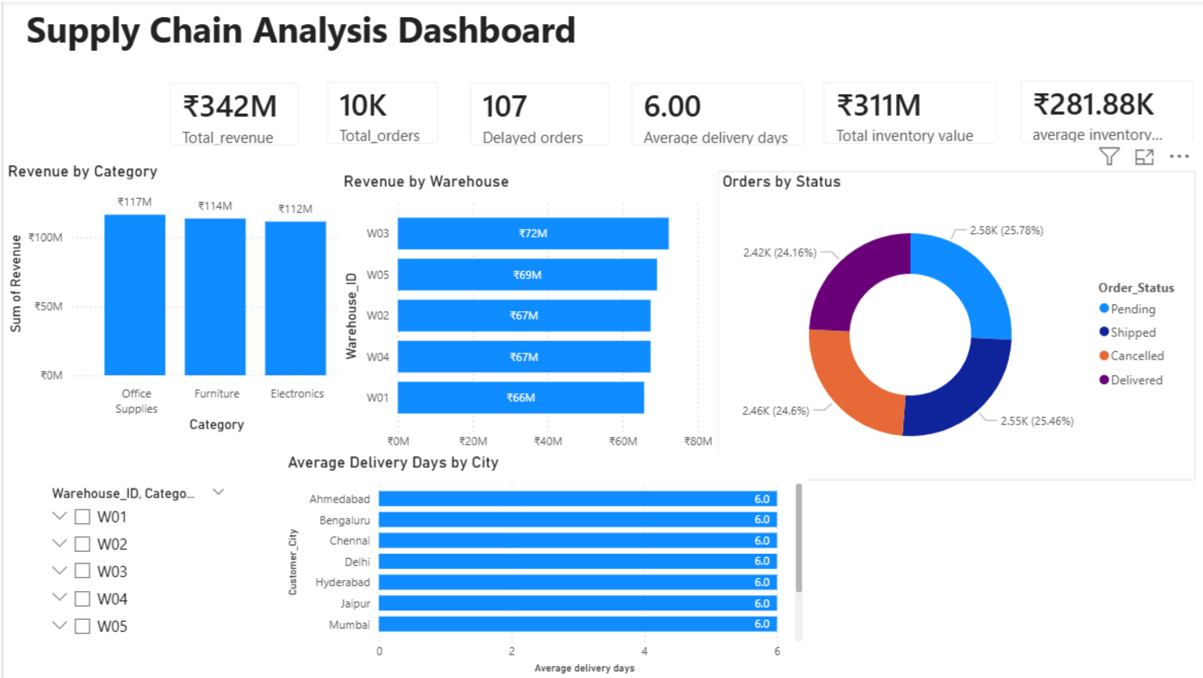
Power BI:



Sheets:

Warehouse_ID	SUM of Revenue
W03	₹72,237,154.00
W05	₹69,109,488.00
W02	₹67,394,109.00
W04	₹67,393,479.00
W01	₹65,681,435.00

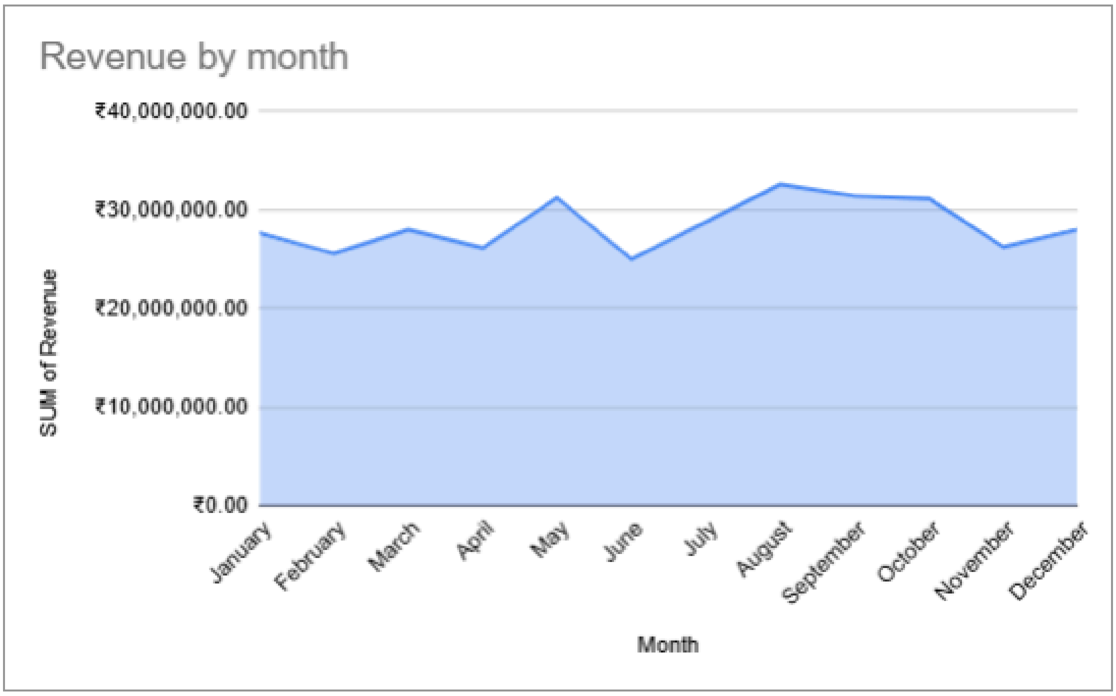
Category	SUM of Revenue
Office Supplies	₹116,539,716.00
Furniture	₹113,744,709.00
Electronics	₹111,531,240.00

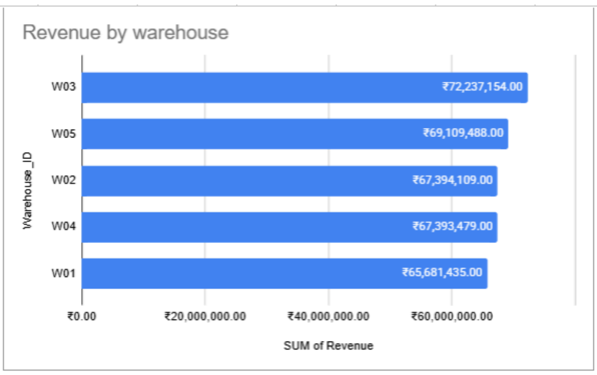
SUM of Revenue	Delayed orders
₹341,815,665.00	107

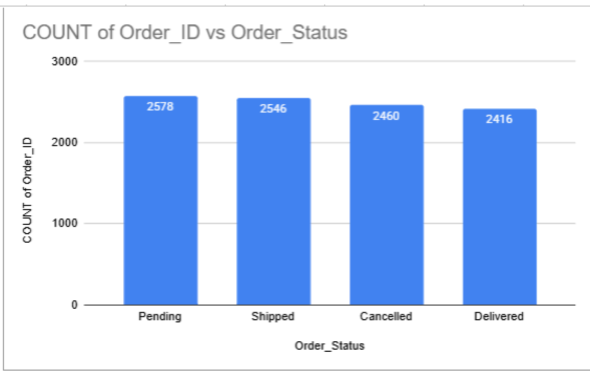
Order_Status	COUNT of Order
Pending	2578
Shipped	2546
Cancelled	2460
Delivered	2416

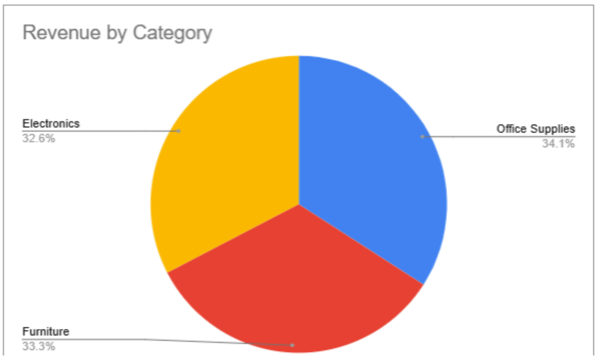
AVERAGE of Revenue
₹34,181.57

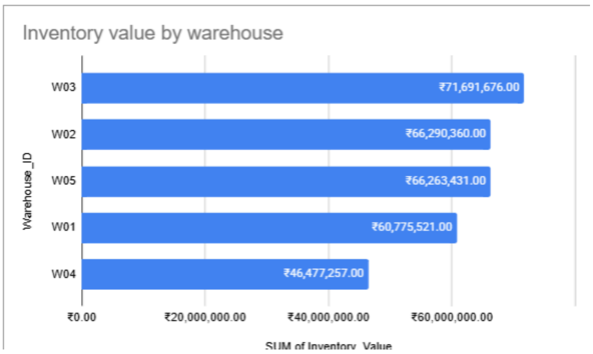
Warehouse_ID	SUM of Inventory
W03	₹71,691,676.00
W02	₹66,290,360.00
W05	₹66,263,431.00
W01	₹60,775,521.00
W04	₹46,477,257.00











Conclusion

This **Supply Chain Analysis** project strengthened my practical skills in **SQL, Excel, and Power BI** by analyzing inventory, warehouse operations, supplier performance, and delivery metrics. The project demonstrated my ability to build dashboards, analyze operational data, and generate actionable insights to support data-driven supply chain decisions.